For Examiner's Use

4 A uniformly charged non-conducting sphere, of radius R, has a positive charge of $+q$ is shown in Fig. 4.1 a. A second uniformly charged non-conducting sphere, which has a negative charge of $-q$, is brought close to the first sphere, such that the distance between their centres is d. The second sphere has a radius r.

Fig. 4.1 a

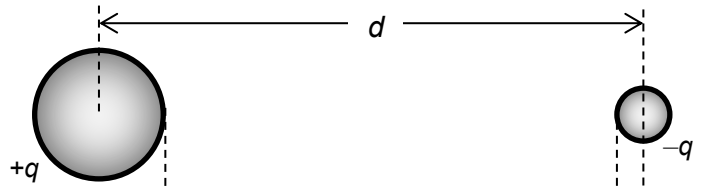


Fig. 4.1 b

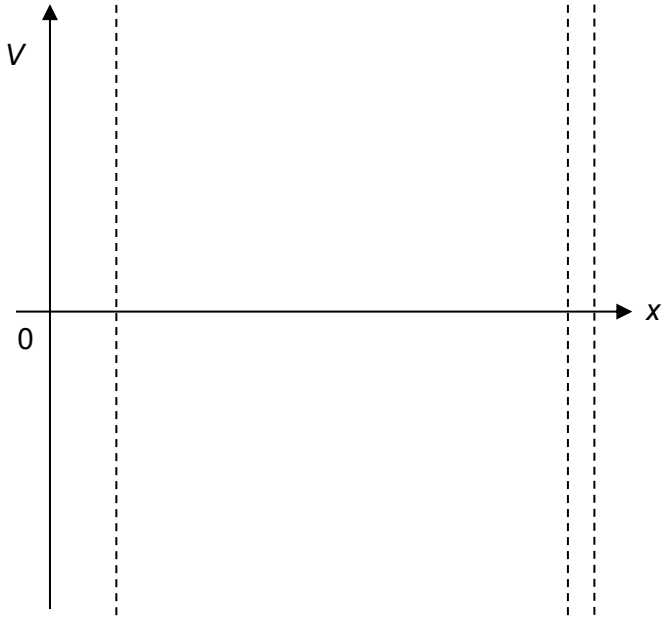
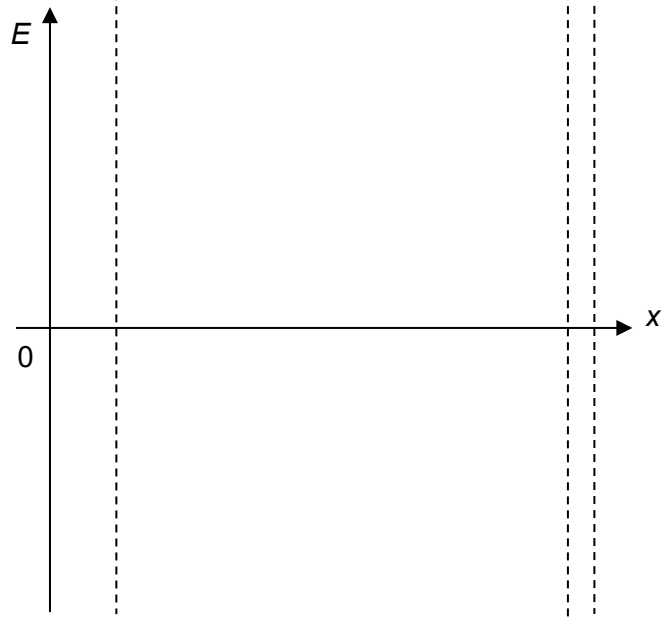


Fig. 4.1 c



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(a) Sketch on Fig. 4.1 b the variation with distance x from the centre of the positively charged sphere of the electric potential V between the surfaces of the charged spheres, due to

(i) the charge on the positively charged sphere only. Label this graph X.

(ii) the charge on the negatively charged sphere only. Label this graph Y.

(iii) the charge on both the positively and negatively charged spheres. Label this graph Z.

[2]

(b) Hence, using Fig. 4.1 b, sketch on Fig. 4.1 c the variation with x of the resultant electric field strength E between the surfaces of the charged spheres.

[2]

(c) The negatively charged sphere is replaced with a charge of $-2q$. A positive point test charge remains stationary when placed at a point along the axis joining the centres of the two spheres.

Determine the distance between this point and the centre of the sphere with $+q$ charge in terms of d.

distance = …… [3]